
Temporally Robust and Fair Credit Risk Prediction

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Abstract

Credit risk models face two structural vulnerabilities in deployment: temporal drift, where year-over-year economic changes degrade predictive accuracy, and demographic bias, where performance optimization concentrates false-positive errors on disadvantaged groups. We train six model families on 200,000 LendingClub loans using a strict temporal split, searching hyperparameters jointly for temporal robustness and worst-case fairness via Optuna TPE. XGBoost achieves the best composite score on the 2016–2017 test set (Final Score 0.600, temporal AUC 0.683), with worst-case income-group FPR disparity of 0.060, far below all competitors, while a post-processing threshold-equalization step reduces residual income disparities by a further 76–92% at negligible performance cost.

1. Introduction

Credit loan evaluation is a high-stakes, high-throughput decision process increasingly delegated to machine learning. Two structural vulnerabilities arise in real deployments that are routinely overlooked when models are selected purely on aggregate accuracy.

Temporal drift (concept drift). The statistical relationship between borrower features and default outcomes shifts year-over-year due to macroeconomic forces: interest rate changes, employment cycles, and evolving lending policies alter what it means for a given loan profile to be risky. A model trained on 2013 data may rank borrowers well within that year’s distribution while failing badly on 2016 loans. Standard random-split evaluation hides this entirely: by shuffling future and past examples into the same training pool, it leaks future distributional information into the model and produces inflated performance estimates.

Demographic bias. Models optimized for aggregate accuracy systematically issue higher false-positive rates to low-income or short-credit-history applicants, even when neither variable is an explicit feature. Proxy features, loan amount, interest rate, FICO score, correlate with income and credit tenure, are some of the included features that

allow the model to discriminate indirectly. The resulting disparate-impact harm is prohibited under the Equal Credit Opportunity Act (ECOA) and the Fair Housing Act, yet optimizing for AUC alone provides no guarantee against it.

The tension between the two. Addressing temporal robustness and fairness simultaneously is non-trivial because regularization choices that reduce concept drift do not necessarily reduce demographic disparity, and vice versa. Models with the highest raw AUC may be the least temporally stable (e.g., LightGBM, AUC drop 0.033 vs. 0.012–0.018 for others) and the least fair, while the most temporally stable model (LR, AUC drop 0.012) can exhibit the worst worst-case income FPR disparity (0.313). A principled framework must optimize both axes jointly.

Research questions. (1) Does jointly penalizing fairness violation in the hyperparameter search objective improve fairness without sacrificing predictive performance relative to a no-skill baseline? (2) Which model family best balances temporal robustness and fairness under this joint objective? (3) Can lightweight post-processing threshold equalization close residual fairness gaps without retraining? (4) How does regularization strength interact with the robustness–fairness tradeoff within each model family?

Related work. Hardt et al. (2016) formalize Equalized Odds and propose post-hoc threshold adjustment as a fairness remedy. We adopt their criterion but move the fairness objective upstream into the hyperparameter search rather than applying it only after training. Calders & Verwer (2010) propose discrimination-free Naive Bayes classifiers that modify the training objective to suppress demographic dependence; our approach is model-agnostic and imposes no constraints on classifier choice. Calmon et al. (2017) take a pre-processing approach, optimizing a transformation of the input data distribution to satisfy fairness constraints while minimizing distortion; we instead encode fairness directly in the hyperparameter search objective with no data modification. Our work extends this line by jointly optimizing temporal robustness and fairness within the hyperparameter search itself, requiring neither data modification, classifier-specific assumptions, nor post-hoc correction, and addressing the temporal stability dimension that prior fairness literature does not consider.

Contributions. We introduce TRP, a year-over-year stabil-

ity metric jointly optimized with worst-case fairness in Optuna, and evaluate six model families under a strict temporal split. XGBoost achieves the best joint performance–fairness tradeoff, and post-processing threshold equalization reduces income FPR disparity by 76–92% at negligible cost.

2. Methods

2.1. Dataset and Preprocessing

We randomly sampled 200,000 records from the public LendingClub loan dataset (random_state=42, issue years 2007–2017). The binary target is loan default, defined as loan status *Charged Off*, *Default*, or *Does not meet the credit policy: Charged Off*. Loans with status *Current* are excluded since their outcome is unknown. The overall default rate is 11.9%, reflecting class imbalance.

2018 exclusion. Loans issued in 2018 are excluded from the test set. Their typical 36–60 month terms had not matured by the dataset’s export date, yielding an artificially low default rate of 1.7% vs. 15.6% for 2016 loans. This is a nine-fold artifact that would corrupt all test metrics.

Features. We use 14 numerical features spanning loan conditions and credit history: `loan_amnt`, `int_rate`, `installment`, `annual_inc`, `dti`, `delinq_2yrs`, `fico_range_low`, `open_acc`, `pub_rec`, `revol_bal`, `revol_util`, `total_acc`, `mort_acc`, `pub_rec_bankruptcies`. Missing values are imputed with training-set medians. StandardScaler is fit on training data only and applied to LR and NN.

Sensitive variables. Two demographic proxies are derived: `income_group` (*low/mid/high*, tercile-split on clipped `annual_inc`) and `credit_group` (*short/mid/long*, tercile-split on credit history length). Both splits use training-data terciles applied consistently to validation and test sets to prevent leakage.

2.2. Temporal Split

We apply a strict forward-looking split: **Train** (issue year ≤ 2013 , $n=20,286$), **Validation** (2014–2015, $n=58,114$), and **Test** (2016–2017, $n=77,616$). All preprocessing statistics (imputation medians, scaler parameters, and fairness group boundaries) are computed on the training partition only and applied to validation and test without refitting.

2.3. Evaluation Metrics

For each model configuration we train *two* versions in parallel: a **temporal model** (fit on data through 2013, evaluated on 2016–2017) and a **random model** (fit on a standard random 70% split of the full dataset, evaluated on the held-out 30%). This dual-training design isolates the cost of concept

drift (Figure 1).

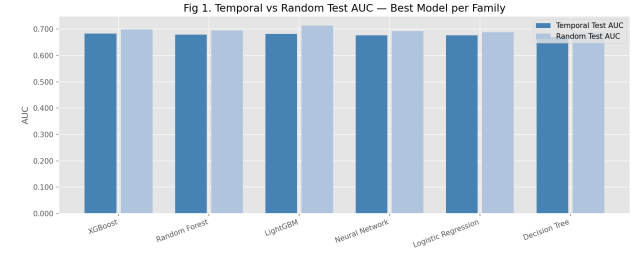


Figure 1. Temporal vs. random-split AUC for each model family. The random model (lighter bars) trains on $\approx 140,000$ shuffled samples; the temporal model (darker bars) trains only on data through 2013 ($n=20,286$) and must generalize across a genuine time gap. The gap between the two bars per model is the Classic AUC Drop.

Classic AUC Drop. The primary answer to our core research question: how much does temporal deployment hurt? is the *Classic AUC Drop*:

$$\Delta\text{AUC} = \text{AUC}_{\text{rand}} - \text{AUC}_{\text{temp}} \quad (1)$$

The difference directly measures performance lost to concept drift: lower is more robust. In our results, LR achieves the smallest drop (0.012) and LGBM the largest (0.033). We note one important limitation: the temporal model trains on $\approx 20,000$ samples (≤ 2013) while the random model trains on $\approx 140,000$ (a 70% random draw from the full 2007–2017 dataset), so the AUC gap conflates concept drift with a $7\times$ training-size disadvantage. Classic AUC Drop is therefore reported as a rough diagnostic of deployment cost, not a pure robustness measure; TRP (below) serves as the primary robustness metric.

Temporally Robust Performance (TRP). The Classic AUC Drop compares two separate models at a single point in time. TRP instead measures *year-by-year stability* of the temporal model alone, penalizing inconsistency across individual calendar years:

$$\text{TRP} = \frac{1}{|T|} \sum_{t \in T} \text{AUC}_t - \lambda \left(\max_t \text{AUC}_t - \min_t \text{AUC}_t \right), \quad \lambda=1.0 \quad (2)$$

where each AUC_t is computed on the temporal model’s predictions for year t only. Setting $\lambda=1$ treats year-over-year variance as equally costly as mean performance. TRP is used directly in the Optuna objective (Eq. (4)); the Classic AUC Drop is reported as a diagnostic alongside it.

Fairness (worst-case threshold sweep). At model selection time, for each model we sweep nine thresholds $\theta \in$

$\{0.1, 0.2, \dots, 0.9\}$ and compute FPR, TPR, and PPR per group in each sensitive variable. The six worst-case scalars (credit FPR/TPR/PPR disparity, income FPR/TPR/PPR disparity) are the maximum over all nine thresholds. This guards against models that appear fair only at one operating point.

Composite Final Score. To rank models on a single axis that reflects both performance and fairness, we define a composite score $S \in (-\infty, 1]$, with higher values indicating better joint performance–fairness. Each term is min-max normalized on the non-baseline model pool (normalization fit on validation, applied to test), so a score of 0 corresponds to the worst model in the pool and 1 to the best on that axis alone:

$$S = 0.35 \text{TRP} + 0.15 \widetilde{\text{AUC}} + 0.10 \text{PR-}\widetilde{\text{AUC}} - 0.40 \bar{F}_6 \quad (3)$$

where $\bar{F}_6$ is the mean of the six min-max normalized worst-case fairness scalars (credit and income group FPR, TPR, and PPR disparity across the nine-threshold sweep). Because higher fairness disparity is *worse*, the penalty term inverts the direction: a model with zero normalized disparity contributes -0.00 to S , while the worst-disparity model contributes -0.40 . The weights reflect domain priorities: TRP (35%) is the primary axis because temporal robustness is the paper’s core claim; the combined performance terms (TRP + AUC + PR-AUC = 60%) are balanced against a 40% fairness penalty, encoding the regulatory requirement that demographic harm cannot be traded away for performance gains. PR-AUC is used instead of accuracy because the 12% positive-class rate makes accuracy uninformative under class imbalance.

2.4. Hyperparameter Optimization

We use Optuna’s Tree-structured Parzen Estimator (TPE, Akiba et al. 2019) with 60 trials per model family. For each trial the threshold is chosen by sweeping $[0.05, 0.95]$ in steps of 0.01 and maximizing validation F_1 . The Optuna objective is then:

$$\text{Optuna Objective} = \text{TRP} - 0.3 \cdot \mathcal{F} \quad (4)$$

where

$$\mathcal{F} = \max(\Delta \text{FPR}^{\text{inc}}, \Delta \text{TPR}^{\text{inc}}, \Delta \text{FPR}^{\text{cred}}, \Delta \text{TPR}^{\text{cred}})$$

is computed at the single F1-optimal threshold on the validation set. Note the distinction: Equation (4) evaluates fairness at one threshold to guide search efficiently; Equation (3) uses the nine-threshold worst-case to select and rank the final models. The top 5 trials per family are retrained and evaluated; the best by composite score is selected.

Models. XGBoost (Chen & Guestrin, 2016), LightGBM (Ke et al., 2017), Random Forest, Neural Network (2-layer

MLP with dropout), Logistic Regression (L1/L2/Elastic-net via Optuna), and Decision Tree are all tuned with the same objective and compared on the same temporal test set.

Baseline. A `DummyClassifier` (`strategy='prior'`) that predicts the training-set class prevalence ($\approx 15.5\%$) for every loan serves as a no-skill lower bound. It achieves $\text{AUC} = 0.500$, $\text{TRP} = 0.500$, $\text{Brier} = 0.108$, $\text{PR-AUC} = 0.122$, and zero fairness disparity, illustrating the paradox that a completely useless model can appear perfectly fair.

2.5. Post-Processing Fairness Intervention

After model selection, we apply per-group threshold equalization: for each demographic group g , we find threshold $\hat{\theta}_g$ on the temporal validation set such that group g ’s FPR matches the overall model FPR at the global threshold. These per-group thresholds are then applied to the test set with no retraining, yielding a model-agnostic final fairness layer.

3. Results

3.1. Test-Set Performance

Table 1 reports all seven models on the held-out test set (2016–2017, $n=77,616$).

Table 1. Test-set results, ranked by Final Score (Eq. (3)). *T-AUC*: temporal-split AUC. *R-AUC*: random-split AUC. *Drop*: $\text{R-AUC} - \text{T-AUC}$ (lower = more robust). *Inc/Cr FPR*: worst-case income/credit FPR disparity across the 9-threshold sweep. Baseline excluded from score normalization (—).

MODEL	SCORE	T-AUC	DROP	TRP	INC FPR	CR FPR
XGB	0.600	0.683	0.016	0.677	0.060	0.022
RF	0.336	0.680	0.016	0.675	0.177	0.078
LGBM	0.185	0.681	0.033	0.676	0.327	0.147
NN	0.014	0.676	0.016	0.668	0.184	0.070
LR	−0.153	0.677	0.012	0.669	0.313	0.163
DTREE	−0.342	0.670	0.018	0.669	0.287	0.164
BASE.	—	0.500	0.000	0.500	0.000	0.000

XGBoost achieves the highest composite score (0.600) by combining competitive TRP (0.677) with by far the lowest worst-case income FPR disparity (0.060) among all models. Its AUC drop of 0.016 is tied for the second-smallest, confirming it is simultaneously temporally stable and robustly fair.

Random Forest ranks second (0.336) with nearly identical temporal AUC (0.680) and TRP (0.675) to XGB, but higher income FPR disparity (0.177) drives its composite score down substantially.

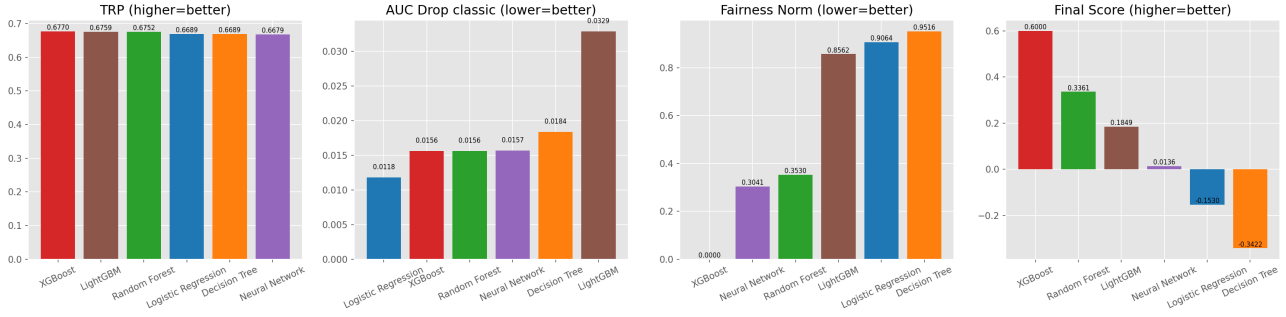


Figure 2. Four-panel test-set comparison across all six model families. **Top-left:** TRP — higher means more stable year-over-year. **Top-right:** Classic AUC Drop — lower means less performance cost from temporal deployment (note: confounded by $7\times$ training-size difference; interpret as rough diagnostic only). **Bottom-left:** Worst-case income FPR disparity across the nine-threshold sweep — lower is fairer. **Bottom-right:** Composite Final Score (Eq. (3)) — higher is better overall. XGB dominates all four panels; LGBM has the best raw AUC but the worst drop and fairness.

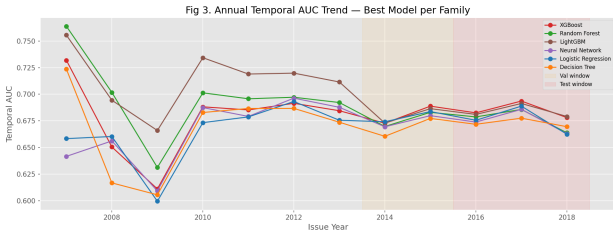


Figure 3. Year-by-year temporal AUC for each model family (2014–2017). The shaded boundary separates the validation window (2014–2015) from the held-out test years (2016–2017). All models are stable across 2014–2016; LGBM shows a notable dip in 2017, while XGB and RF remain the most consistent. TRP penalizes this year-over-year variance, which single-number AUC masks entirely.

LightGBM achieves the highest raw random-split AUC (0.714) but pays with an AUC drop of 0.033, which is more than double every other model, while the the worst income FPR disparity (0.327). High raw discriminability comes at the cost of temporal fragility and demographic harm.

Neural Network scores just above zero (0.014) with competitive AUC (0.676) and moderate fairness (income FPR 0.184). Its lower PR-AUC (0.213) relative to tree-based models reflects a weaker ability to identify the minority default class.

Logistic Regression has the smallest AUC drop (0.012), making it the most temporally *stable* model by that diagnostic. However, its income FPR disparity reaches 0.313 under worst-case threshold conditions, exposing that temporal stability and fairness are not the same property and do not travel together.

Decision Tree scores lowest (−0.342), penalized by the

weakest PR-AUC (0.197) and high fairness disparities (income FPR 0.287, credit FPR 0.164). Its poor performance on the minority default class makes it unsuitable for this imbalanced credit-risk task.

All real models substantially outperform the baseline on every axis: AUC 0.67–0.68 vs. 0.50, TRP 0.67–0.68 vs. 0.50, and PR-AUC 0.20–0.22 vs. 0.12 (equal to class prevalence, the theoretical no-skill ceiling for PR-AUC).

Figure 2 visualizes TRP, AUC drop, worst-case income FPR disparity, and composite score side-by-side across all models, confirming XGB’s dominance on all four axes simultaneously. Figure 3 shows year-by-year AUC trajectories, making the concept drift visible: LGBM trends inconsistently across years, with a notable dip in 2017, yet its aggregate AUC Drop (0.033) is more than double every other model, illustrating that AUC Drop and TRP capture complementary aspects of temporal behavior.

Table 2. Ablation endpoint values (validation set). *LR*: C = inverse regularization strength. *XGB*: ss = subsample fraction. *RF*: depth = max tree depth. FPR disp. = worst-case income FPR disparity.

MODEL	PARAM	AUC	TRP	FPR DISP.
LR	$C=10^{-4}$	0.500	0.500	0.000
LR	$C=0.01$	0.682	0.662	0.292
LR	$C=100$	0.678	0.669	0.334
XGB	$ss = 0.4$	0.683	0.664	0.067
XGB	$ss = 1.0$	0.681	0.663	0.043
RF	DEPTH = 3	0.682	0.663	0.250
RF	DEPTH = 15	0.669	0.654	0.325

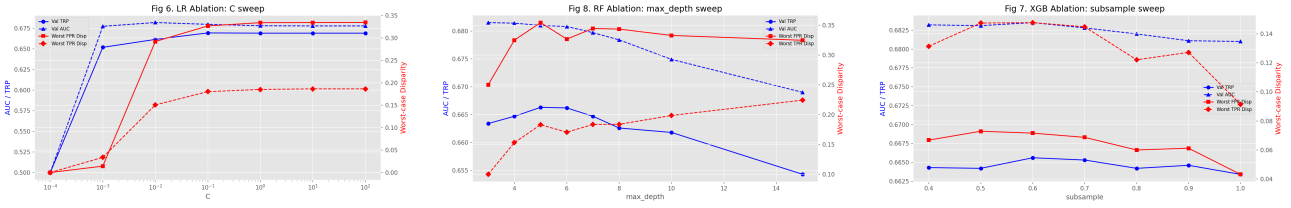


Figure 4. Ablation studies (validation set). **Left:** LR regularization cliff. At $C=10^{-4}$ the model collapses to random (AUC 0.500, disparity 0.000); recovering discriminability at $C=0.01$ immediately triggers high income FPR disparity (0.292). No intermediate region exists where LR is both accurate and fair. **Center:** RF max-depth sweep. Shallower trees achieve higher AUC, higher TRP, and lower income FPR disparity simultaneously. **Right:** XGB subsample sweep. AUC is nearly flat across subsample fractions 0.4–1.0 (range 0.003); FPR disparity is lowest at subsample 1.0.

3.2. Ablation Study

We fix each family’s best Optuna parameters and sweep one key regularization hyperparameter to isolate how regularization strength interacts with the robustness–fairness tradeoff.

LR regularization cliff. At $C=10^{-4}$, LR collapses to random (AUC 0.500) but is trivially fair (disparity 0.000): identical to the DummyClassifier. At $C=0.01$, discriminability recovers sharply (AUC 0.682) but income FPR disparity *explodes* from near-zero to 0.292. Above $C=0.01$, both AUC and disparity plateau regardless of further relaxation. There is no intermediate operating region where LR is both accurate and fair.

RF depth tradeoff. AUC degrades monotonically from depth 3 (0.682) to depth 15 (0.669), while worst-case FPR disparity is lowest at depth 3 (0.250) and peaks around depth 5 (0.354). Shallower trees are simultaneously better on AUC, TRP, and fairness. Regularization here reduces both overfitting and proxy-variable exploitation simultaneously.

XGB subsample. AUC is largely insensitive to the subsample fraction (range 0.003 across 0.4–1.0). FPR disparity is highest at subsample 0.5 (0.073) and lowest at subsample 1.0 (0.043), non-monotonically. XGB’s composite-score dominance is therefore structural rather than threshold-sensitive.

Table 2 summarizes the endpoint values for each sweep.

3.3. Fairness Intervention

After selecting the best model per family, we apply per-group threshold equalization and evaluate the change in FPR disparity on the test set. Figure 5 shows the negligible AUC cost of the intervention; Figure 6 shows the before-and-after FPR disparity for each model and demographic axis.

Income-group FPR disparity is reduced by 83% for XGB (0.224→0.037), 92% for LR (0.194→0.016), and 76% for RF (0.180→0.044). These “before” values are measured at

each model’s F1-optimal operating threshold (≈ 0.16 – 0.17), which falls between the discrete points of the nine-threshold sweep and therefore may differ from the worst-case values reported in Table 1. Credit-group improvements are similarly large (60–85%). The negligible performance cost confirms that residual fairness gaps stem primarily from miscalibrated operating thresholds, not from biased probability scores embedded in the model weights. Threshold equalization is therefore an effective and inexpensive final layer deployable on any of the six model families.

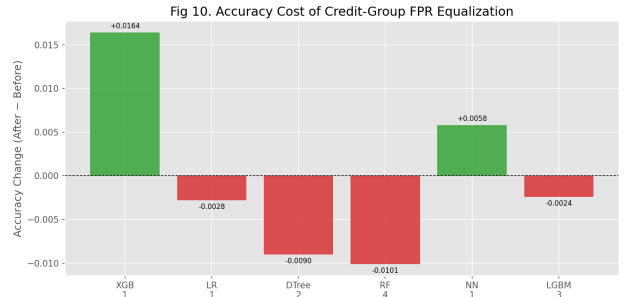


Figure 5. Performance cost of threshold equalization (test set). Each model is shown before (circle) and after (cross) per-group threshold adjustment. AUC decreases are at most 0.003 across all models, confirming that fairness gains come at negligible predictive cost.

4. Process

4.1. Evolution of the Temporal Robustness Metric

We tried two earlier TRP formulations before the final one.

V1: Classic AUC Drop (retained as diagnostic). We trained two separate models — one on a random split, one on a strict temporal split — and measured the AUC gap (Eq. (1)). This directly answers “how much does temporal deployment hurt?”: positive values mean concept drift costs the model discriminability it would have had under ideal random-split conditions. We retain this metric as a reported diagnostic. However, AUC Drop alone is insufficient as a

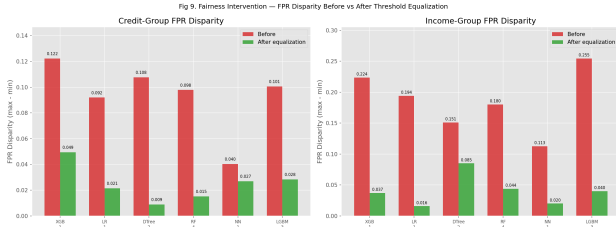


Figure 6. Post-processing threshold equalization results (test set). Each pair of bars shows worst-case FPR disparity before (light) and after (dark) per-group threshold adjustment, for income groups (top) and credit groups (bottom). Income disparities fall 76–92% across all models; credit disparities fall 60–85%. Performance cost is negligible ($\Delta\text{AUC} \leq 0.003$).

robustness metric: a model can have a small AUC Drop yet collapse in one specific year and recover in another, measuring aggregate drift cost but not year-over-year *stability*.

V2: Dual-evaluation AUC difference (abandoned). We trained a single temporal-split model and evaluated it on both a random hold-out and a temporal hold-out, comparing the two AUC scores. This was discarded because the gap reflects distributional differences between the two test sets, not year-over-year decay of the same model, and cannot be used as a search signal inside Optuna.

V3 (final): Year-level TRP. Equation (2) measures AUC variance *within* a single temporal evaluation environment. By computing AUC separately for each calendar year and penalizing the range, TRP captures year-over-year stability directly. It complements Classic AUC Drop: AUC Drop quantifies the cost of concept drift relative to a random-split baseline; TRP quantifies how consistent the temporal model is across individual deployment years.

4.2. Objective Function Evolution

Our initial Optuna objective was a weighted sum of TRP, F_1 , and a fairness penalty. Realized ranges differed sharply (AUC: 0.67–0.68; fairness: 0.10–0.37), causing fairness to dominate; MinMax normalization fixed this but violates Optuna’s trial-independence assumption. Removing F_1 , since AUC already integrates recall–precision balance, left all performance metrics unchanged, yielding the simplified objective $\text{TRP} - 0.3 \cdot \mathcal{F}$, which needs no normalization and is stable trial-to-trial.

4.3. 2018 Data-Maturity Artifact

Initial experiments included 2018 loans in the test set. We discovered that the 2018 default rate (1.7%) was a factor of nine lower than 2016 (15.6%): loans issued in 2018 had not yet reached their 36–60 month maturity by the data export date, so most defaults were simply not recorded. Including

2018 in the test set caused all models to look artificially well-calibrated and inflated accuracy and AUC. Excluding 2018 restores a realistic evaluation distribution.

4.4. LR Calibration Fix

Setting `class_weight='balanced'` in the LR objective artificially up-weighted default-class predictions, inflating predicted probabilities for the minority class and producing a Brier score of 0.229, which is worse than the DummyClassifier (0.108). Removing this flag corrected Brier to 0.104, restoring meaningful probability calibration and enabling fair comparison across models.

4.5. Redefining Fairness Metrics: PPR and the Optuna Objective

We initially considered FPR, TPR, and PPR (Positive Prediction Rate / Statistical Parity) as disparity metrics, but all three cannot be simultaneously satisfied — the fairness impossibility theorem. In credit lending, PPR equalizes loan *approval rates* regardless of repayment ability, conflicting with the goal of lending only to applicants likely to repay. We therefore use Equalized Odds (FPR and TPR only) as the Optuna fairness criterion (Eq. (4)); PPR disparity is retained in $\bar{F}_6$ for monitoring but does not drive hyperparameter selection.

5. Contributions

Implementation. (1) We designed and implemented TRP (Eq. (2)) as a trainable signal in the Optuna hyperparameter search, penalizing year-over-year AUC variance alongside a worst-case fairness penalty. (2) We implemented the nine-threshold worst-case fairness sweep, computing FPR, TPR, and PPR disparity across income and credit-history groups at nine operating thresholds to prevent cherry-picked fairness. (3) We conducted ablation studies for LR (C), XGB (subsample), and RF (max_depth), exposing a sharp regularization–fairness cliff in LR and a monotone fairness–depth tradeoff in RF. (4) We implemented post-processing per-group threshold equalization, reducing income FPR disparity by 76–92% at negligible performance cost. Code: [GitHub Repository](#) and Drive: [Google Drive](#)

Data. We used the public LendingClub loan dataset (2007–2017), 200,000 records, available at <https://www.kaggle.com/datasets/wordsforthewise/lending-club>. Income and credit-history group labels are derived from training-set tercile boundaries.

Libraries. Optuna (Akiba et al., 2019), scikit-learn (Pedregosa et al., 2011), XGBoost (Chen & Guestrin, 2016), LightGBM (Ke et al., 2017), pandas, numpy, matplotlib.

6. AI Usage

We used Claude (Anthropic) and ChatGPT (OpenAI) for targeted assistance across several stages of the project. For ideation and experimental design, AI helped brainstorm evaluation approaches and fairness metric choices, though all final decisions were made by the team. For coding, AI assisted with Python debugging and resolving implementation errors. For writing, AI helped with LaTeX formatting, figure placement, and iterative refinement of report text. AI was not used for hyperparameter tuning, result interpretation, or analysis. All of those were done independently by the team. All code was written and executed by us, and all numbers, figures, and claims in this report are our own.

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